

Inferring Sleep Deprivation Risk Patterns from NHANES Data

Identifying Key Drivers of Sleep Deprivation

Dae Young Kim, Jihoon Choi, Victor Pietono
STT 811: Applied Statistical Modeling

Abstract

Sleep deprivation is a widespread yet underdiagnosed public health issue, linked to chronic illness, impaired cognition, and reduced quality of life. Using nationally representative NHANES data, we developed predictive models to flag individuals at risk of sleep deprivation based on demographic, behavioral, and clinical variables. Our pipeline involved dual-stage feature selection and multiple modeling approaches including rule-based classifiers, Generalized Linear Models (GLMs), XGBoost, neural networks, and a stacking ensemble. The best-performing GLM (BIC-based) achieved an AUC of 0.76 and F1 of 0.51, while ensemble methods yielded slightly higher recall. Key predictors included bedtime, smoking status, race/ethnicity, and sedentary behavior, with interaction effects (e.g., bedtime $\times$ gender) revealing important subgroup differences. The results support the use of routine health data to build scalable, interpretable tools for early screening and intervention in sleep health.

1 Introduction

Sleep deprivation is a widespread yet underdiagnosed public health concern associated with increased risks for chronic conditions such as diabetes, cardiovascular disease, and mental health disorders. It also significantly impairs cognitive function, productivity, and overall quality of life. A major challenge in detecting sleep deprivation arises because its symptoms, like fatigue and irritability, closely mimic other medical or psychological conditions, complicating timely diagnosis in routine healthcare settings.

Traditional approaches for identifying sleep deprivation largely depend on self-reported sleep questionnaires, which are not consistently integrated into clinical visits and often lack broader contextual data. While these self-reports offer valuable insights, their scope and accuracy are inherently limited. To address this critical gap, our study leverages comprehensive health, demographic, and lifestyle data alongside sleep questionnaires. Using nationally representative data from the National Health and Nutrition Examination Survey (NHANES), we aim to identify the key determinants of sleep deprivation and develop interpretable, scalable predictive models. This approach will facilitate early screening of individuals at risk, thereby supporting proactive intervention strategies and enhancing awareness of sleep health in both clinical and public health contexts.

2 Data and Methods

We used data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative health survey administered by the Centers for Disease Control and Prevention (CDC). NHANES integrates interviews, physical examinations, laboratory tests, and questionnaires, offering a comprehensive view of health conditions and behaviors across the U.S. population.

For this study, we combined data from the 2017–2020 and 2021–2023 cycles, resulting in a pooled sample of approximately 18,000 individuals. The dataset captures diverse information, including laboratory biomarkers related to inflammation, metabolism, lipids, hormones, and nutrition. Demographic variables such as age, gender, race/ethnicity, and educational attainment are included, along with physical measurements like weight, height, body mass index (BMI), waist circumference, and blood pressure. Questionnaire data further enrich the dataset with responses on sleep habits, smoking and alcohol use, income, and weight history. The multidimensional scope of NHANES provides an ideal foundation for predictive modeling efforts aimed at identifying individuals at risk of sleep deprivation.

2.1 Target Variable Construction

Our target variable, sleep deprivation status, was constructed from self-reported sleep timing data by calculating a weighted average of weekday and weekend sleep durations. Participants averaging fewer than 7 hours of sleep per day were classified as sleep-deprived. To prevent potential data leakage, sleep end times were excluded from the predictor variables, with sleep start times and contextual variables utilized instead.

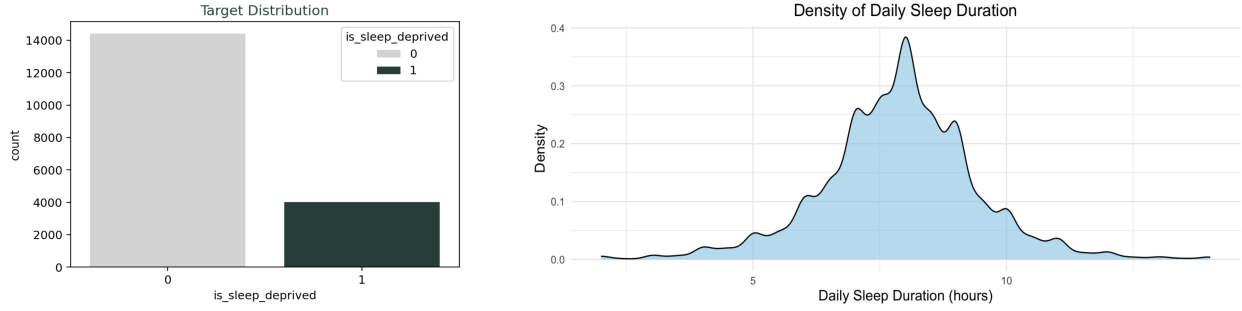


Figure 1: Distribution of sleep deprivation status and daily sleep duration

The resulting classification exhibited notable class imbalance, with approximately 22% of individuals identified as sleep-deprived. To address this imbalance and ensure robust model performance, we employed weighting and oversampling techniques during model training. Analysis of the continuous sleep duration variable indicated a roughly normal distribution centered around 7 to 8 hours, with a pronounced tail toward shorter sleep durations reflecting sleep-deprived individuals. This distribution further supported our clinically relevant classification threshold of 7 hours.

2.2 Data Cleaning and Preprocessing

We recoded non-informative responses (e.g., "Refused," "Don't know") and blanks as missing. Variables with over 20% missingness, near-zero variance, or missing target values were dropped. The dataset included both numeric and categorical variables with a high degree of heterogeneity, so we split them accordingly for imputation.

Numeric features were imputed using an XGBoost-based regressor, while categorical features were imputed using an XGBoost classifier through an iterative imputation process. For models requiring interpretability—particularly the Generalized Linear Model (GLM)—we carefully addressed multicollinearity using Variance Inflation Factor (VIF) testing, applying a conservative threshold of 5. Multicollinearity can compromise GLM performance by causing parameter instability, inflating standard errors, and undermining the interpretability of coefficients. To mitigate these risks, variables with high VIF were removed one at a time, and VIF was recalculated after each removal to account for shifts in the correlation structure among remaining predictors. While modern machine learning models like ensemble trees are more robust to collinearity, rigorous collinearity control was necessary for GLM to preserve its explanatory value.

2.3 Feature Selection

To identify the most relevant predictors of sleep deprivation, we employed a multi-step feature selection strategy combining both statistical and model-based approaches.

First, we applied *filter methods* to the cleaned dataset. For numeric variables, we used ANOVA F-tests to identify the top 30 features most strongly associated with the target. For categorical variables, Chi-squared tests were used to select the top 10 features. This filter-based approach helped isolate features with strong univariate relationships to sleep deprivation risk.

Next, we implemented a *wrapper method* using Recursive Feature Elimination (RFE) with an XGBoost classifier. This approach evaluated the predictive contribution of each feature in the context of the full model and selected the top 40 features through iterative removal of less informative variables.

The features selected from both filter and wrapper methods were then combined into a single set. To ensure model parsimony, we further reduced this list to the top 30 features based on their importance scores from a fitted XGBoost model. This final feature set balanced statistical relevance, predictive utility, and interpretability, serving as the foundation for all subsequent modeling.

2.4 Exploratory Data Analysis

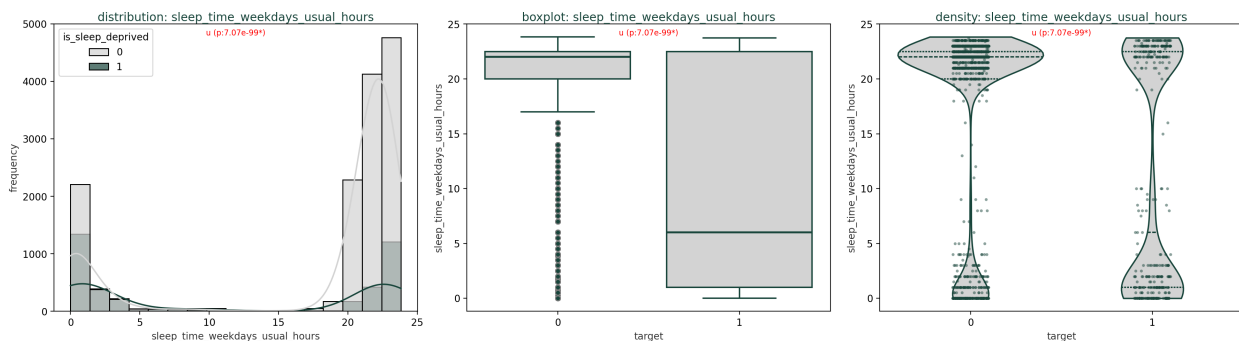


Figure 2: Key numerical predictor: Bedtime

Exploratory Data Analysis (EDA) focused on identifying key predictors of sleep deprivation from both numerical and categorical variables. For numerical features—including bedtime, age, smoking status, and waist-to-height ratio—we examined data distributions, medians, densities, and outliers through histograms, box plots, and violin and strip plots. We tested for normality using Shapiro-Wilk tests; features with normal distributions were analyzed using T-tests to compare means between sleep-deprived (Class 1) and non-deprived (Class

0) groups. Non-normally distributed features underwent Mann-Whitney U tests to compare medians. This analysis highlighted bedtime, age, and smoking status as particularly significant numerical predictors.

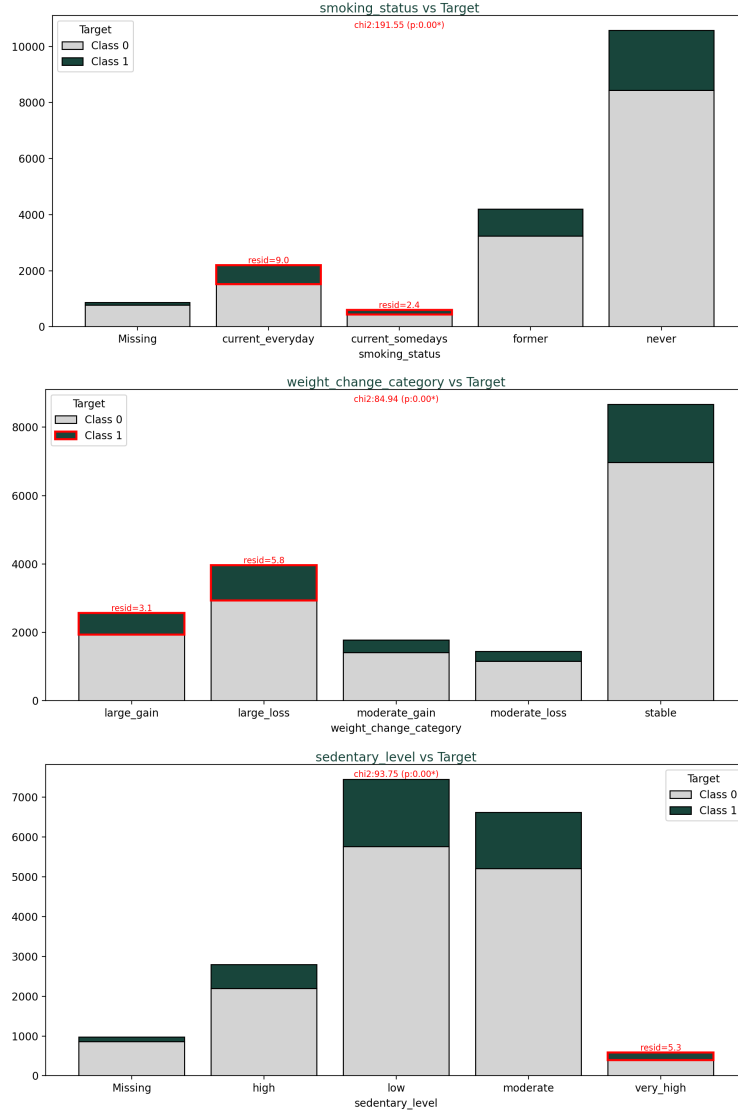


Figure 3: Key categorical predictors: Smoking status, weight change, sedentary level

Categorical variables were evaluated using Chi-squared tests to determine their overall association with sleep deprivation. To further explore these relationships, we reviewed standardized residuals, identifying which categories specifically drove the significant associations. Notable categorical predictors included current smoking status, weight-change status, and a very high sedentary lifestyle, all strongly associated with increased sleep deprivation risk.

These significant findings from the EDA informed feature selection and guided the construction of our baseline predictive model.

2.5 Feature Engineering

We introduced additional features to enhance our model’s predictive capabilities. First, a global continuous feature—waist-to-height ratio—was calculated for all models based on the initial EDA on measurement dataset.

Specifically for generalized linear models (GLMs), which require careful feature engineering to effectively capture nonlinear patterns, we created a binary variable named *weekday_sleep_timing*. This feature categorizes sleep start times into two risk groups: low-risk (18:00–23:00) and high-risk (00:00–17:00). To further support GLMs in capturing complex relationships, we included interaction terms between this binary sleep timing variable and key demographic variables such as race, gender, and education level.

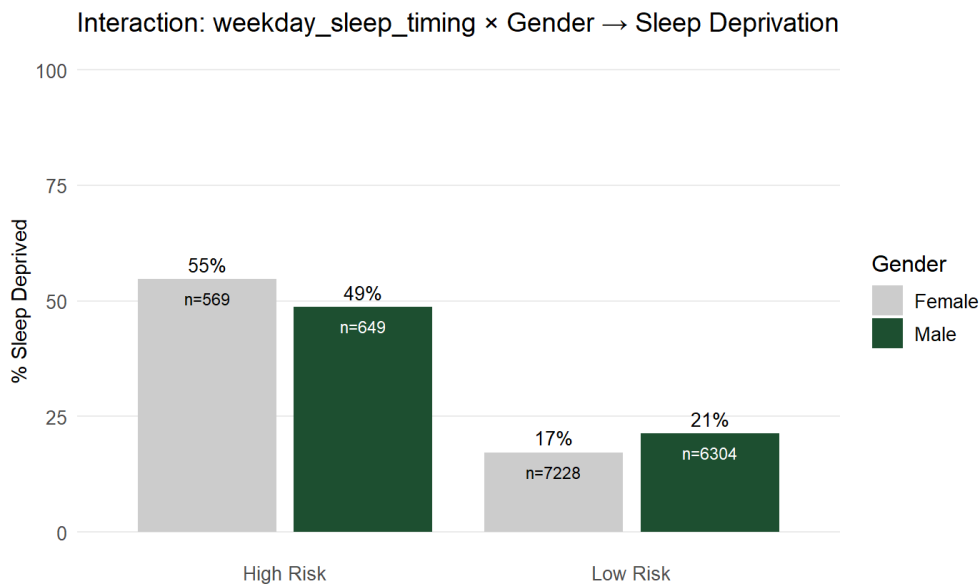


Figure 4: The split bar chart shows crossing pattern (females higher in one category, lower in the other) suggests an interaction between these variables

To break the non-linearity in the key variable *weekday_sleep_timing*, we applied a binarization and put it as categorical variable. Additionally, to ensure bedtime did not inadvertently introduce data leakage, we locked the binning procedure (24-hour intervals) for sleep start times to uniformly applied across training and testing datasets by rounding to nearest hour. The data from test will not affecting the binning decision.

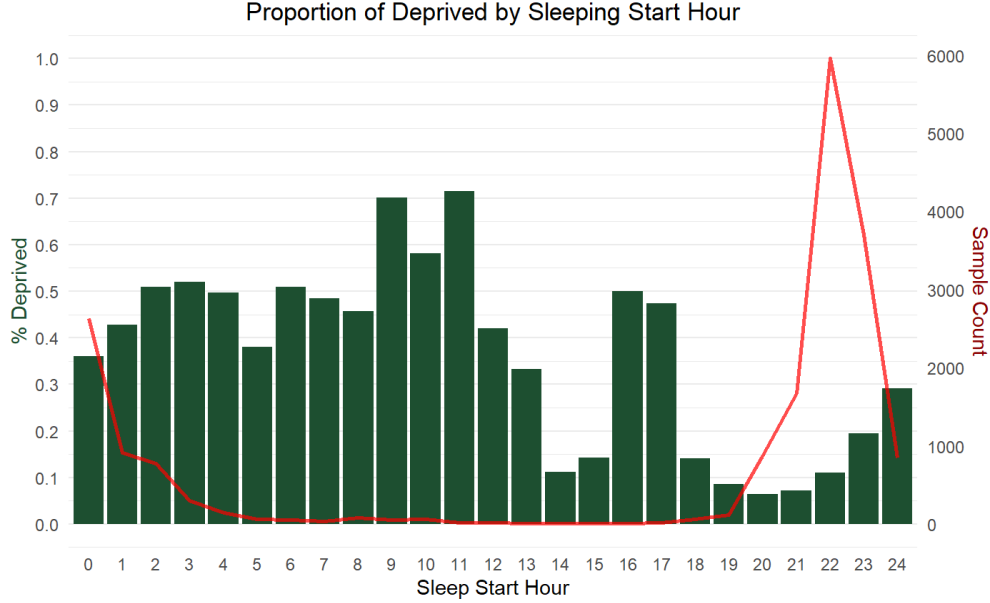


Figure 5: The distribution of sleep deprived based on sleep timing suggest a complex non-linear relationships

3 Model Construction

We explored a variety of modeling techniques to effectively classify sleep deprivation risk, starting from a straightforward, rule-based classifier and progressing toward more sophisticated approaches.

3.1 Naive Model

The baseline was a simple, interpretable model informed directly by exploratory data analysis (EDA). It identified individuals as sleep-deprived if at least two of the following conditions were met: bedtime between midnight and 8 AM, age between 30 and 60, current smoking status, or very high sedentary behavior. This basic approach provided a benchmark, achieving a modest AUC of 0.65.

3.2 XGBoost

Next, we implemented gradient boosting using XGBoost, addressing the notable class imbalance (approximately 22% sleep-deprived cases) by adjusting the class weights to enhance sensitivity toward the minority class. We carefully tuned hyperparameters—including tree depth, learning rate, number of estimators, and class weights—using manual grid search combined with 5-fold cross-validation. XGBoost significantly outperformed the baseline model,

attaining an AUC of 0.77, recall of 0.69, and F1-score of 0.52, effectively capturing most cases critical to health screening contexts.

3.3 Neural Networks

Recognizing that neural networks can detect subtle, higher-order interactions among predictors, we trained a Multi-layer Perceptron (MLP). We utilized a single hidden-layer architecture, exploring various hidden layer sizes (e.g., 50, 100 units), activation functions (ReLU, tanh), and L2 regularization strengths. Class imbalance was managed through random over-sampling, with hyperparameters optimized using grid search and cross-validation. While the MLP demonstrated good ability to model complex relationships, performance (AUC = 0.69, F1 = 0.47) was somewhat below XGBoost.

3.4 Stacked Model

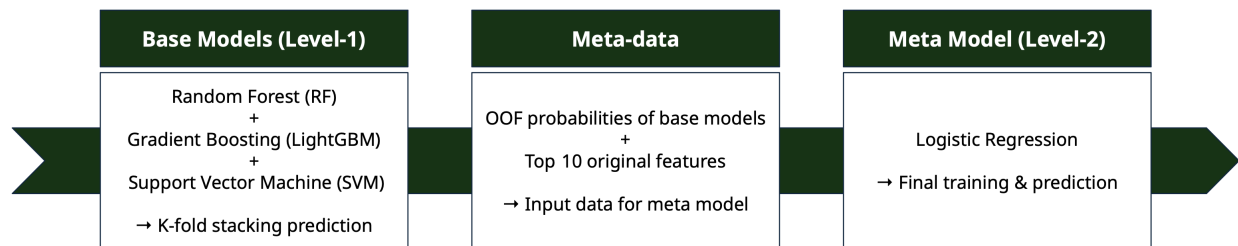


Figure 6: Stacked ensemble architecture for sleep deprivation prediction

We also constructed a stacked model designed to leverage the complementary strengths of diverse algorithms. This approach combined three distinct base learners—Random Forest for robustness against noise, LightGBM for efficiency and handling imbalance, and SVM for boundary-sensitive classification. These base models generated out-of-fold predicted probabilities (meta-data), which were then augmented with the top 10 original predictive features. The combined features were scaled for stability and fed into a logistic regression meta-model for final training and prediction. This ensemble achieved performance comparable to XGBoost, with an AUC of 0.77, recall of 0.70, and F1-score of 0.51.

Each model was rigorously evaluated through 5-fold cross-validation, allowing us to compare trade-offs in complexity, interpretability, and predictive accuracy comprehensively.

3.5 Generalized Linear Model

Since this is a healthcare problem where interpretation is important, we implemented GLM as one of our potential models. We used a binomial distribution with a logistic link function,

reserving non-canonical links only if diagnostics indicated necessity.

For our GLM baseline, we ran an untuned model using all available features without any feature engineering, achieving a test AUC of 0.73, Sensitivity of 0.63, and F1-score of 0.47.

Unlike tree-based or neural network models, GLMs require explicit feature engineering to model nonlinearities and interactions. Our tuned GLM model incorporates the previously described engineered features, including the weekday_sleep_timing categorical variable and its demographic interactions.

We then explored two approaches to GLM optimization: stepwise selection and regularization. For stepwise selection, we implemented both AIC and BIC criteria in forward, backward, and forward-backward directions. For regularization, we tested LASSO, Ridge, and Elastic Net models. Table 1 presents the performance comparison across all methods.

BIC-selected models showed performance comparable to AIC models but with fewer features (12 variables for BIC Forward vs. 22 for AIC Forward). For better interpretability and parsimony, we selected the BIC Forward model as our final GLM. It achieved a test AUC of 0.76, sensitivity of 0.70, and F1-score of 0.51.

While regularized models achieved similar predictive performance, they introduced shrinkage bias, making direct coefficient interpretation invalid under the maximum likelihood framework. Therefore, despite their utility to boost the model performance, the drawback for interpretation purpose specific in our case doesn't justify compared to BIC Forward model.

Table 1: GLM model performance comparison across selection and regularization methods.

Model	# Variables	# Features	Test AUC	Test Sensitivity	Test F1	Train F1
GLM (Untuned)	28	54	0.73	0.63	0.47	0.48
AIC Forward	22	67	0.77	0.71	0.51	0.51
AIC Backward	22	68	0.77	0.71	0.51	0.51
AIC Both	19	64	0.76	0.71	0.51	0.51
BIC Forward	12	51	0.76	0.70	0.51	0.51
BIC Backward	14	54	0.76	0.70	0.51	0.51
BIC Both	12	51	0.76	0.70	0.51	0.51
LASSO*	NA	59	0.76	0.71	0.52	0.51
Ridge*	NA	77	0.76	0.71	0.51	0.51
Elastic Net*	NA	59	0.76	0.71	0.52	0.51

*Regularization introduces shrinkage bias in the estimates and violates the MLE property, making direct coefficient interpretation invalid.

4 Model Selection and Validation

4.1 Model Comparison and Selection

We evaluated multiple modeling approaches to identify sleep deprivation risk. The baseline rule-based model, derived from exploratory analysis conditions (bedtime between 12:00am-8:00am, age 30-60, current smoker, very high sedentary level), achieved an AUC of 0.65, recall of 0.44, and F1-score of 0.41.

Generalized Linear Models demonstrated strong performance with appropriate tuning. Using BIC-forward selection, the GLM achieved an AUC of 0.76, recall of 0.70, and F1-score of 0.51. Among machine learning models, XGBoost achieved the highest performance (AUC: 0.77, recall: 0.69, F1-score: 0.52), while a stacked model combining Random Forest, LightGBM, and SVM models reached similar metrics (AUC: 0.77, recall: 0.70, F1: 0.51). The neural network performed moderately (AUC: 0.69, recall: 0.63, F1: 0.47).

Table 2: Model performance comparison on the test set.

Model	Test AUC	Test Sensitivity	Test F1 Score
Naive Model (Baseline)	0.65	0.44	0.41
GLM (Untuned)	0.72	0.63	0.47
GLM (BIC Forward)	0.76	0.70	0.51
Boosting Ensemble (XGBoost)	0.77	0.69	0.52
Stacking Ensemble (RF + LightGBM + SVM)	0.77	0.70	0.51
Neural Net (MLP)	0.69	0.63	0.47

Given its balance of performance and interpretability, we selected the BIC-forward GLM as our final model. A finely-tuned GLM offered comparable predictive capability to more complex models while maintaining transparency for risk factor communication in context of public health.

4.2 Model Diagnostics

After selection, we performed comprehensive diagnostic checks on the BIC-forward GLM. The deviance residuals plot shows the expected symmetric split pattern typical for binary outcome model, with the smoothing line remains close to zero across the predicted values. This confirms the appropriateness of the binomial distribution and logit function.

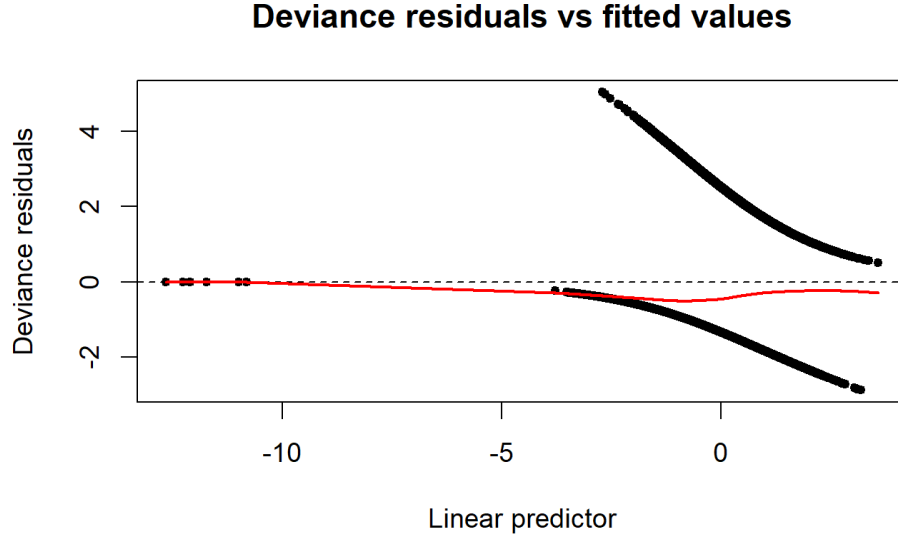


Figure 7: Deviance residuals vs fitted values for the BIC Forward GLM model. The symmetric pattern and smooth line near zero confirm appropriate distribution and link function selection.

Cross-validation confirmed the model’s stability across five folds with consistent performance. The minimal difference between training and test performance further supports the model’s stability and there is no indication of overfitting.

Table 3: Five-fold cross-validation results for GLM (BIC Forward).

Fold	Sensitivity	F1 Score	AUC
Fold 1	0.70	0.52	0.76
Fold 2	0.67	0.48	0.77
Fold 3	0.66	0.50	0.76
Fold 4	0.71	0.52	0.77
Fold 5	0.70	0.51	0.77
Average $\pm$ SD	0.69 ± 0.02	0.51 ± 0.01	0.77 ± 0.01

Table 4: GLM (BIC Forward) performance on training and test data.

Dataset	Sensitivity	F1 Score	AUC
Train Data	0.69	0.51	0.77
Test Data	0.70	0.51	0.76

5 Results

Key predictors identified consistently across models included sleep start time (bedtime), health behaviors, and demographic factors. Sleep timing was particularly influential: individuals starting sleep early (between 18:00 and 23:00) had significantly reduced odds of sleep deprivation, with the strongest protective effects occurring between 19:00 and 21:00 (peak protection around 20:00).

Regarding health behaviors, tobacco use notably increased sleep deprivation risk, whereas individuals experiencing stable or moderate weight changes had lower risk. Demographically, males and college graduates exhibited lower odds of being sleep-deprived. Conversely, Black individuals displayed higher odds of sleep deprivation compared to other racial groups, indicating potential underlying disparities that warrant further investigation. Additionally, an interaction between sleep timing and gender emerged: bedtime affected males and females differently, suggesting nuanced gender-specific sleep patterns influencing sleep deprivation risk.

Table 5: Key predictors of sleep deprivation from BIC Forward GLM model

Predictor	Estimate	Std. Error	z value	p-value
<i>Sleep Timing</i>				
Sleep at 18:00	-1.918	0.237	-8.102	<0.001***
Sleep at 19:00	-2.251	0.197	-11.416	<0.001***
Sleep at 20:00	-2.396	0.085	-28.103	<0.001***
Sleep at 21:00	-2.170	0.064	-34.170	<0.001***
Sleep at 22:00	-1.582	0.041	-38.853	<0.001***
Sleep at 23:00	-0.889	0.042	-21.246	<0.001***
<i>Health Behaviors</i>				
Tobacco use	0.217	0.036	6.059	<0.001***
Weight change (moderate gain)	-0.223	0.056	-3.977	<0.001***
Weight change (moderate loss)	-0.259	0.060	-4.336	<0.001***
Weight change (stable)	-0.251	0.042	-6.023	<0.001***
Fatigue flag	0.242	0.043	5.648	<0.001***
<i>Demographics</i>				
Gender (male)	-0.337	0.086	-3.897	<0.001***
Education (college graduate)	-0.342	0.053	-6.420	<0.001***
Race (Black)	0.406	0.053	7.666	<0.001***
Race (White)	-0.203	0.050	-4.058	<0.001***
<i>Interaction Effects</i>				
Low risk sleep timing $\times$ Male	0.562	0.090	6.215	<0.001***

Significance codes: '***' 0.001, '**' 0.01, '*' 0.05, '.' 0.1

6 Conclusion

This study demonstrates the potential to predict sleep deprivation risk using routinely collected demographic, clinical, and behavioral data, eliminating the need for dedicated sleep questionnaires. Our modeling approach showed that interpretable models such as Generalized Linear Models (GLMs) can perform competitively with more complex machine learning techniques, offering a practical balance between transparency and predictive accuracy. Boosting and ensemble methods, while slightly enhancing recall, captured more intricate nonlinear patterns, especially beneficial in identifying high-risk individuals.

Among all predictors, sleep start time (bedtime) emerged as the most influential factor, with early sleep onset—particularly between 19:00 and 21:00—strongly associated with reduced risk. Additional key predictors included tobacco use, race/ethnicity, weight change

history, and sedentary behavior level. The presence of interaction effects, such as the differing impact of bedtime on males and females, highlights the complex and individualized nature of sleep deprivation risk.

These findings provide a foundation for developing scalable, low-cost screening tools that can be integrated into routine health assessments. With further validation and the application of causal inference methods, these models could support more timely identification and targeted interventions for individuals at elevated risk of sleep deprivation—improving both clinical care and population health outcomes.

7 Discussion

Despite promising results, this study presents several limitations. The use of self-reported sleep timing introduces the potential for recall bias and misclassification, which may affect the validity of the target variable. Additionally, the observational nature of NHANES data constrains the findings to associations rather than causal inferences.

Future research should incorporate causal inference frameworks such as propensity score matching, weighting, or structural causal models to better isolate the effects of individual predictors. Further subgroup analyses stratified by occupational status, income level, and mental health condition could illuminate how sleep deprivation risk varies across contexts and help refine the application of predictive tools for different populations.

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